

Zomato Bangalore

Restaurant & Ratings Analysis

Data cleaning, exploratory insights, and platform recommendations

Prepared for an Alfido Tech-style analytics platform
Source: zomato.csv (Bangalore restaurant listings)

Executive Summary

This report analyses restaurant listings scraped from Zomato for Bangalore, covering ratings, cuisines, locations, pricing, and service features (online ordering, table booking). The raw file contained a significant data-quality problem — roughly half of all rows had review text bleeding into structured columns — which was detected and cleaned before analysis. The resulting dataset of just over 10,000 unique restaurants shows clear patterns: certain premium central neighbourhoods and mid-to-niche cuisines (Continental, Cafe, Desserts, Asian) consistently out-rate the high-volume categories (North Indian, Fast Food, Biryani); price and table-booking availability are the strongest correlates of a higher rating; and restaurant supply is heavily concentrated in a handful of outer-Bangalore micro-markets that don't necessarily rate the best. Five recommendations for a food-discovery / analytics platform are provided at the end of this report.

1. Data Cleaning

1.1 The corruption problem

The source CSV had 13 well-formed columns per row structurally (every row parsed to exactly 13 fields), but for a large share of records the actual *values* were shifted — fields such as **rate**, **location**, or **listed_in(type)** contained fragments of restaurant review text rather than the values those columns should hold. This is an upstream data-generation issue, not a parsing artifact.

Each structural column was validated against its expected pattern (e.g. **rate** should match *4.1/5*, *NEW*, -, or be blank; **online_order/book_table** should be *Yes/No*; **listed_in(type)** should be one of seven known categories). Rows failing any check were treated as unrecoverable and dropped rather than guessed at.

Step	Rows
Raw rows	56,252
Structurally valid (kept)	29,549 (52.5%)
After removing exact duplicate listings	10,296

1.2 Field-level cleaning

- **Rating:** extracted numeric value from 'X.X/5' strings; 'NEW' and '-' treated as missing.
- **Votes:** converted to integer.
- **Cost for two:** stripped thousands-separator commas and converted to numeric Rs. value (already in Indian Rupees — no currency conversion required).
- **Text fields** (name, location, rest_type, cuisines, dish_liked): whitespace-trimmed; empty/'nan' strings standardised to missing.

- **Duplicates:** exact duplicate restaurant listings (same name + address, arising because a restaurant can appear once per service category) were dropped, leaving one row per unique restaurant.

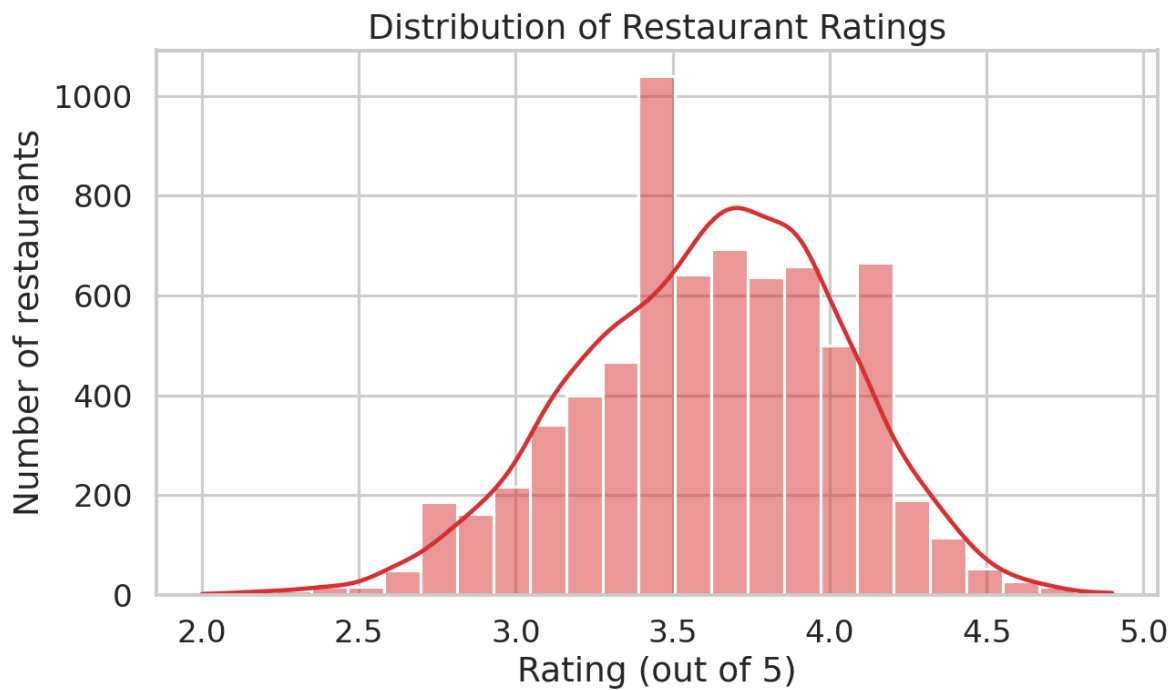
1.3 Feature engineering

- **primary_cuisine / primary_rest_type** — first cuisine / restaurant-type tag from each comma-separated list.
- **n_cuisines** — count of cuisines offered.
- **cost_bucket** — price bracket (less than Rs.300, Rs.300–600, Rs.600–1000, Rs.1000–1500, Rs.1500–2500, Rs.2500+).

2. Exploratory Analysis

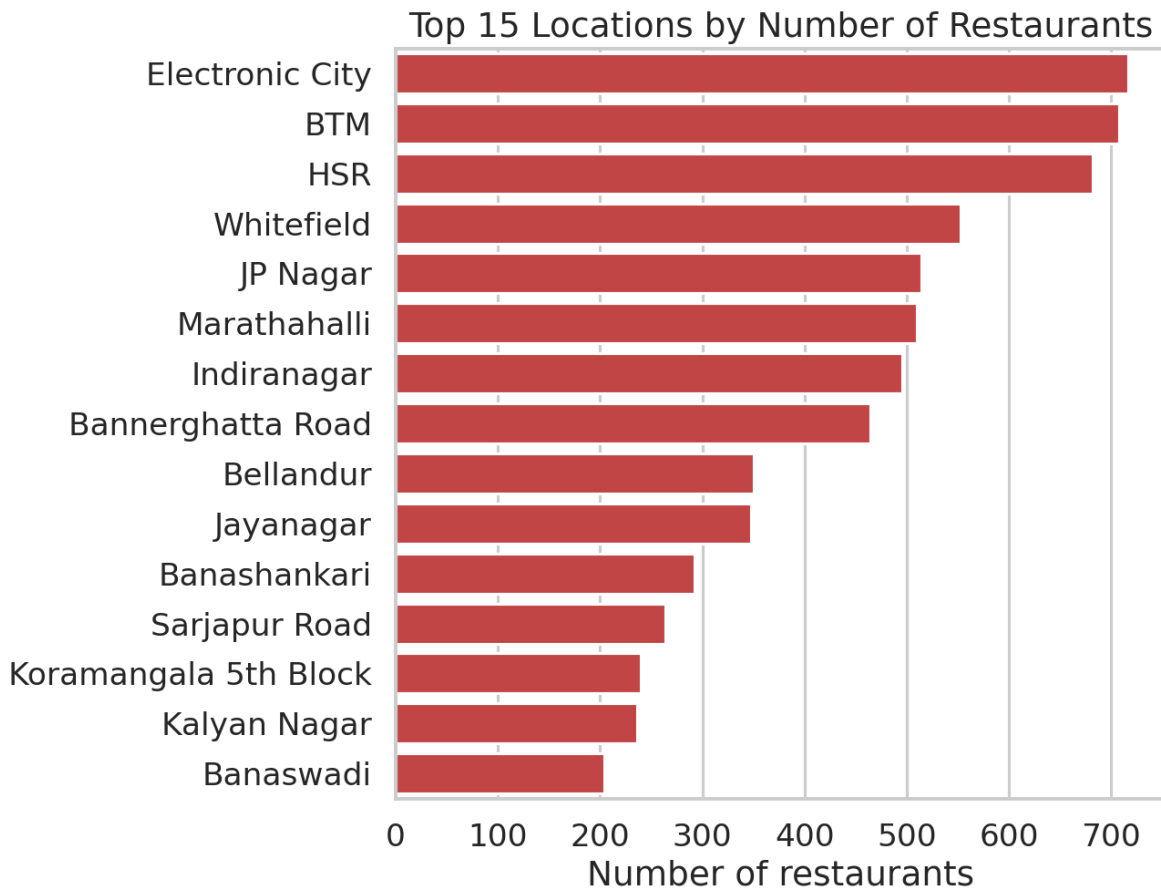
2.1 Rating distribution

Across 10,296 cleaned restaurant listings, the average rating is **3.62 / 5**. Ratings cluster tightly between 3.0 and 4.3 — very few restaurants score below 2.5 or above 4.5, suggesting rating inflation/compression typical of platform review systems.



2.2 Location hotspots

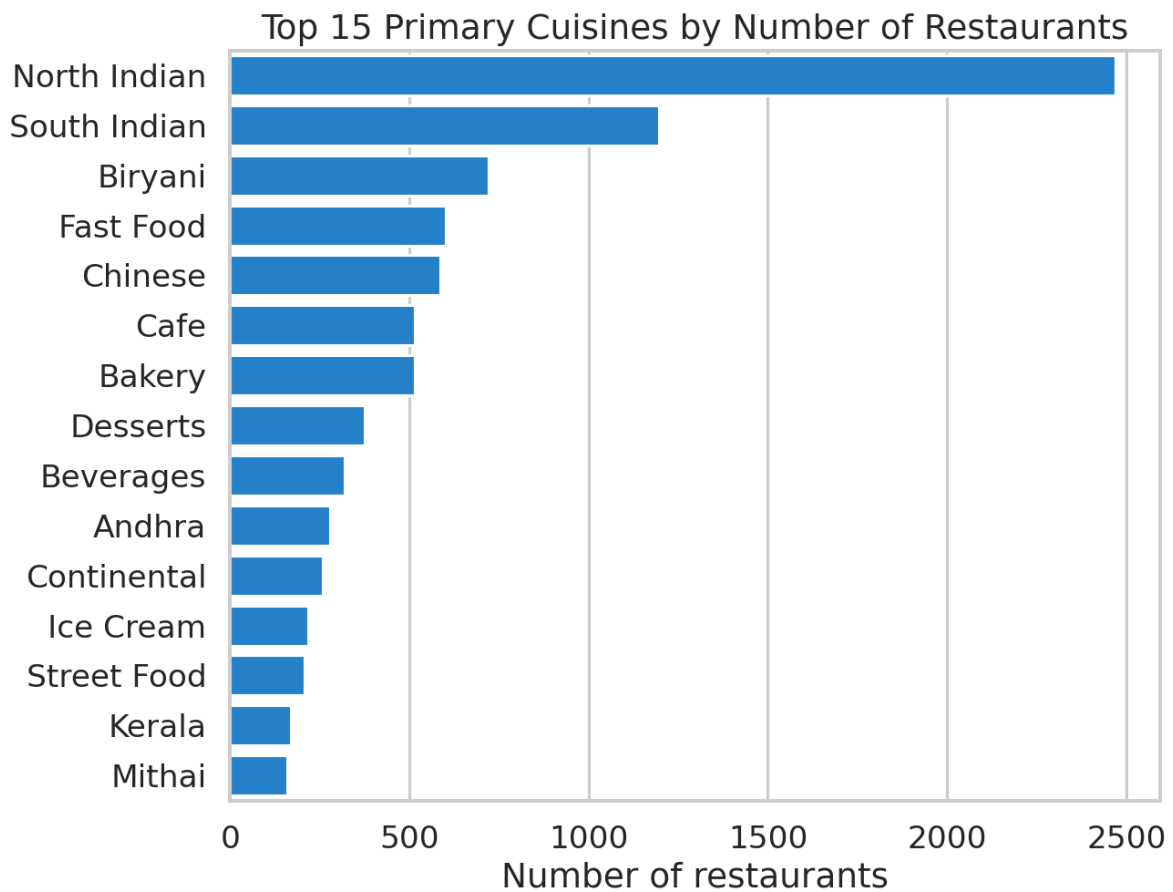
Restaurant supply is heavily concentrated in a handful of outer-Bangalore neighbourhoods — Electronic City, BTM, HSR, Whitefield and JP Nagar together account for a large share of all listings.



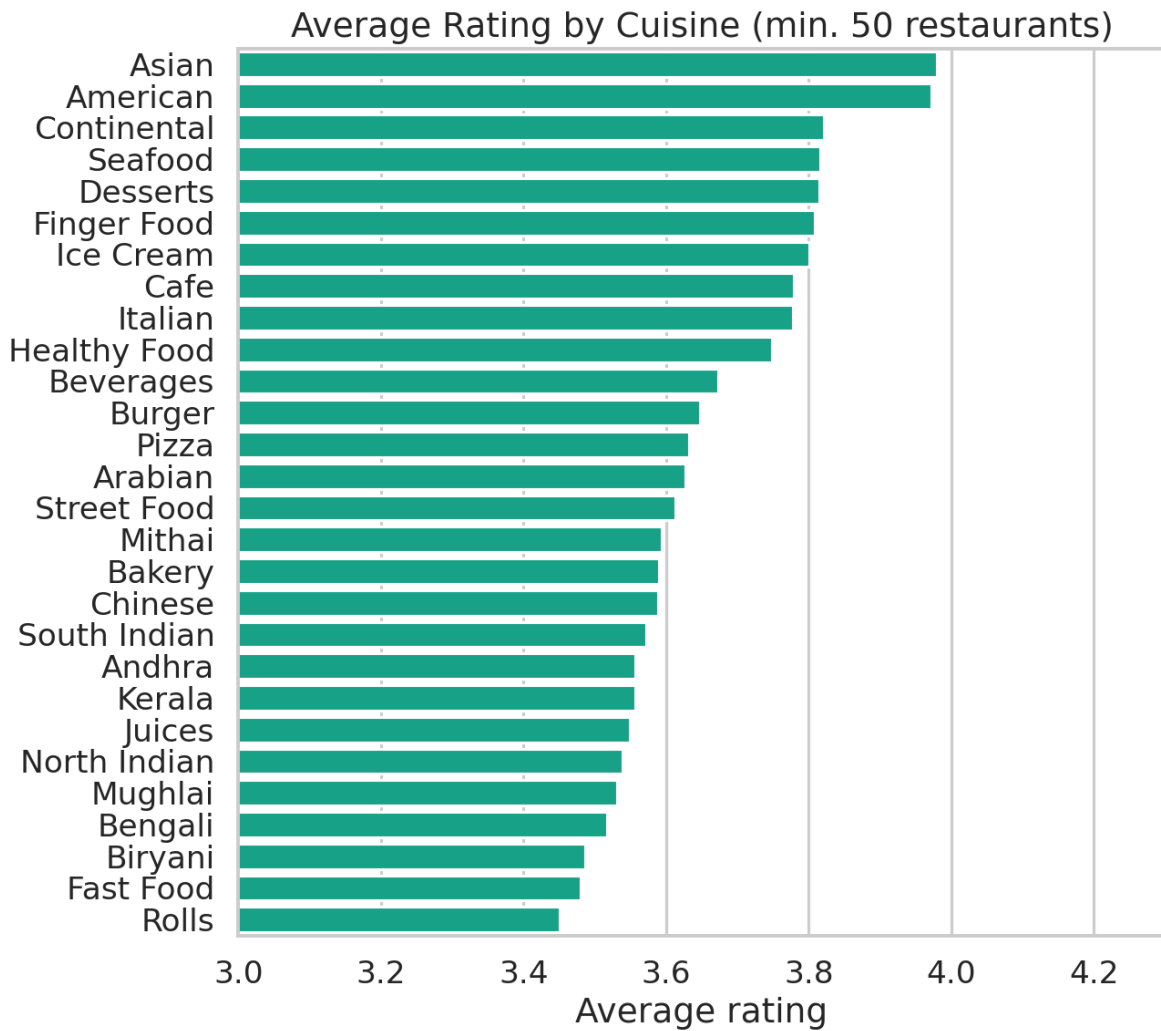
However, the highest *average* ratings are found in more central, premium areas rather than the highest-supply ones: Lavelle Road (4.04), St. Marks Road (3.93), Church Street (3.93), and Koramangala 5th Block (3.88) lead among locations with at least 30 restaurants — while high-volume Electronic City and Marathahalli average closer to 3.4–3.5 in cuisine-specific cuts (see heatmap, section 2.5).

2.3 Cuisine popularity vs. quality

North Indian, South Indian, Biryani, Fast Food and Chinese dominate by sheer restaurant count.



But popularity and rating quality diverge: niche/premium cuisines rate noticeably higher on average than the high-volume staples.

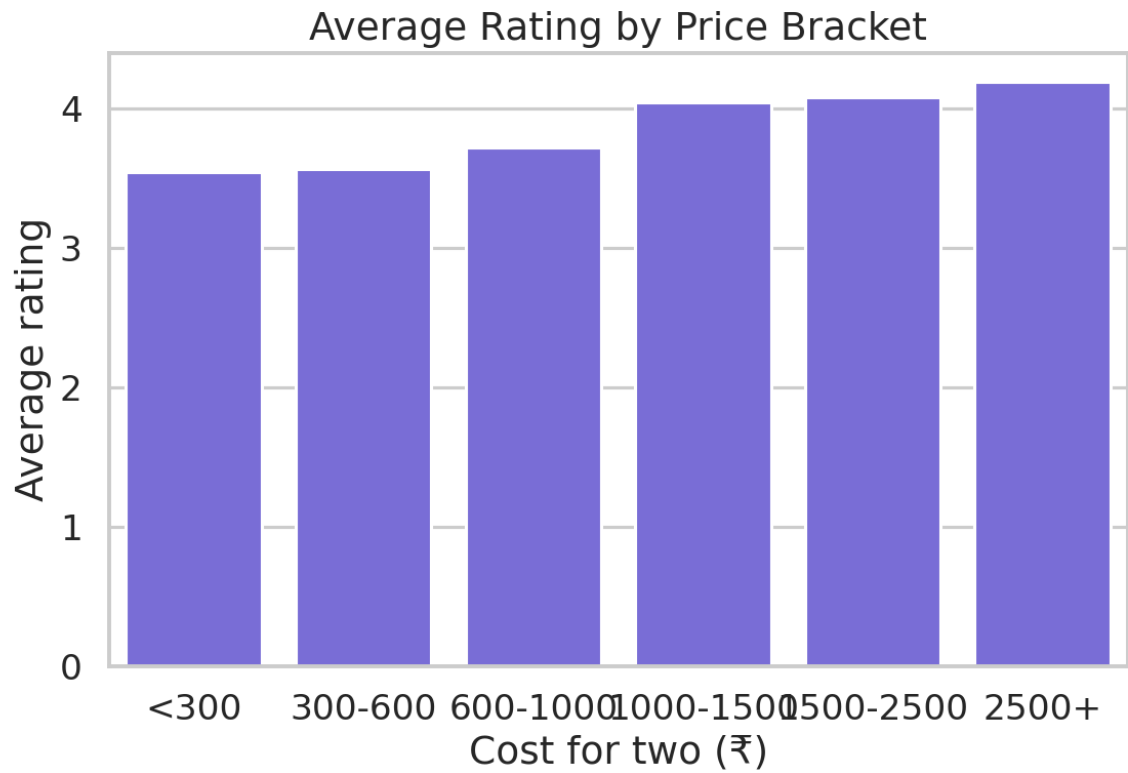


Asian (3.98), American (3.97) and Continental (3.82) lead; Biryani (3.49), Fast Food (3.48) and Rolls (3.45) trail, despite far higher restaurant counts.

2.4 Price vs. rating

Cost for two and rating are positively correlated (Pearson $r \approx 0.30$) — pricier restaurants tend to be rated higher, likely reflecting better service, ambience, and food consistency at higher price points.

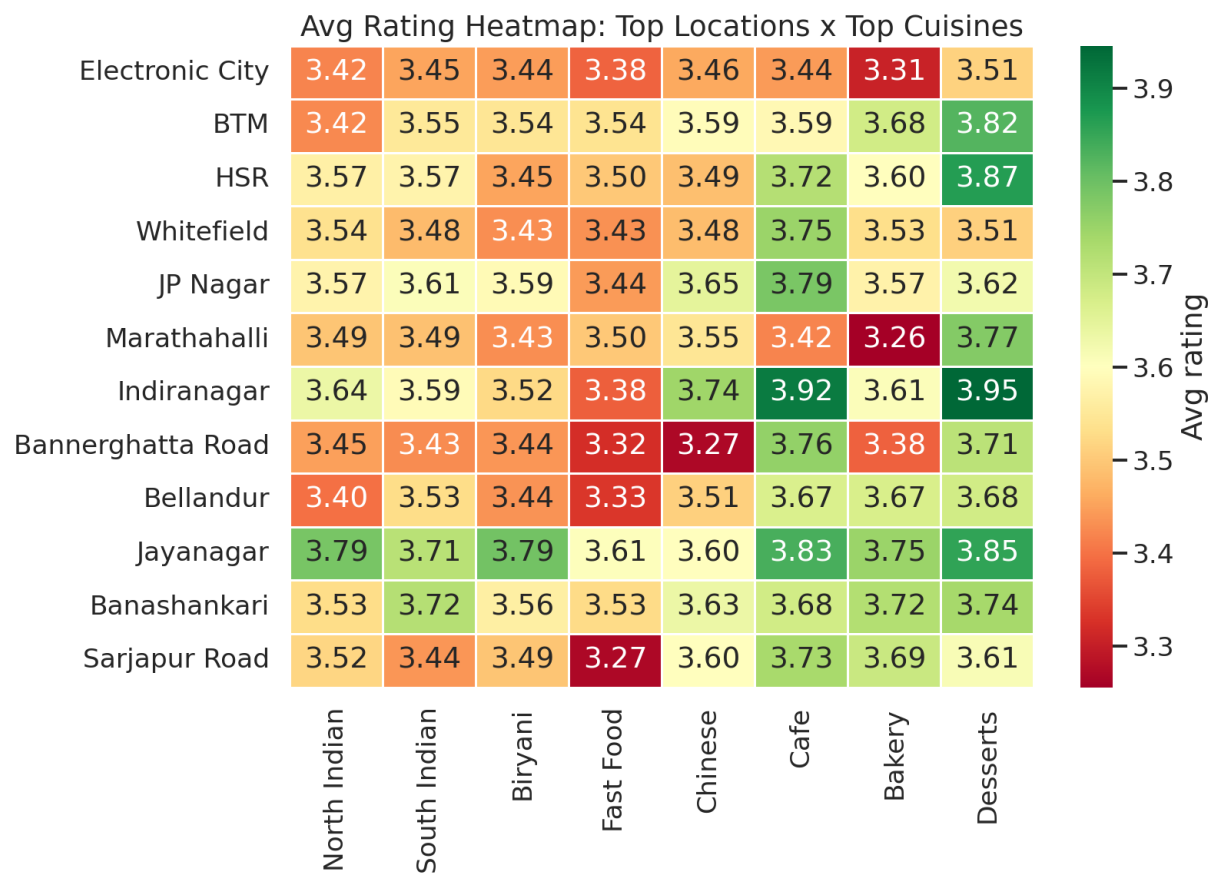




Average rating rises steadily with price bracket, from 3.55 (under Rs.300) to 4.19 (Rs.2500+ for two).

2.5 Location × cuisine hotspot map

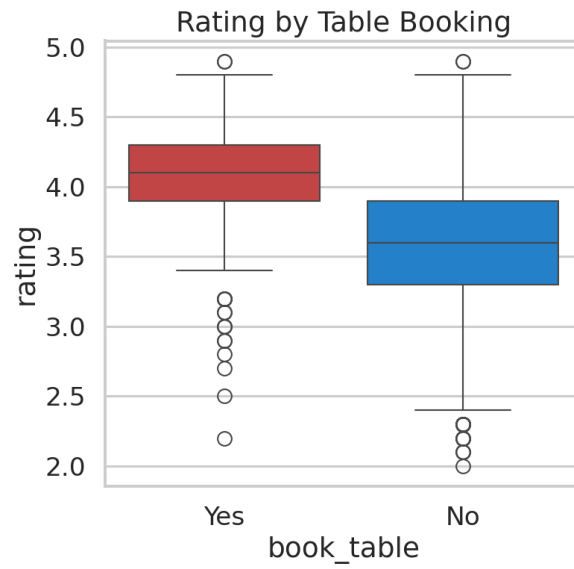
Cross-referencing the 12 highest-supply locations against the 8 most common cuisines reveals where specific cuisine types under- or out-perform locally — useful for both restaurant partnerships and consumer guidance.



Indiranagar and Jayanagar rate strongly across most cuisines; Fast Food underperforms almost everywhere, most sharply in Sarjapur Road and Bannerghatta Road.

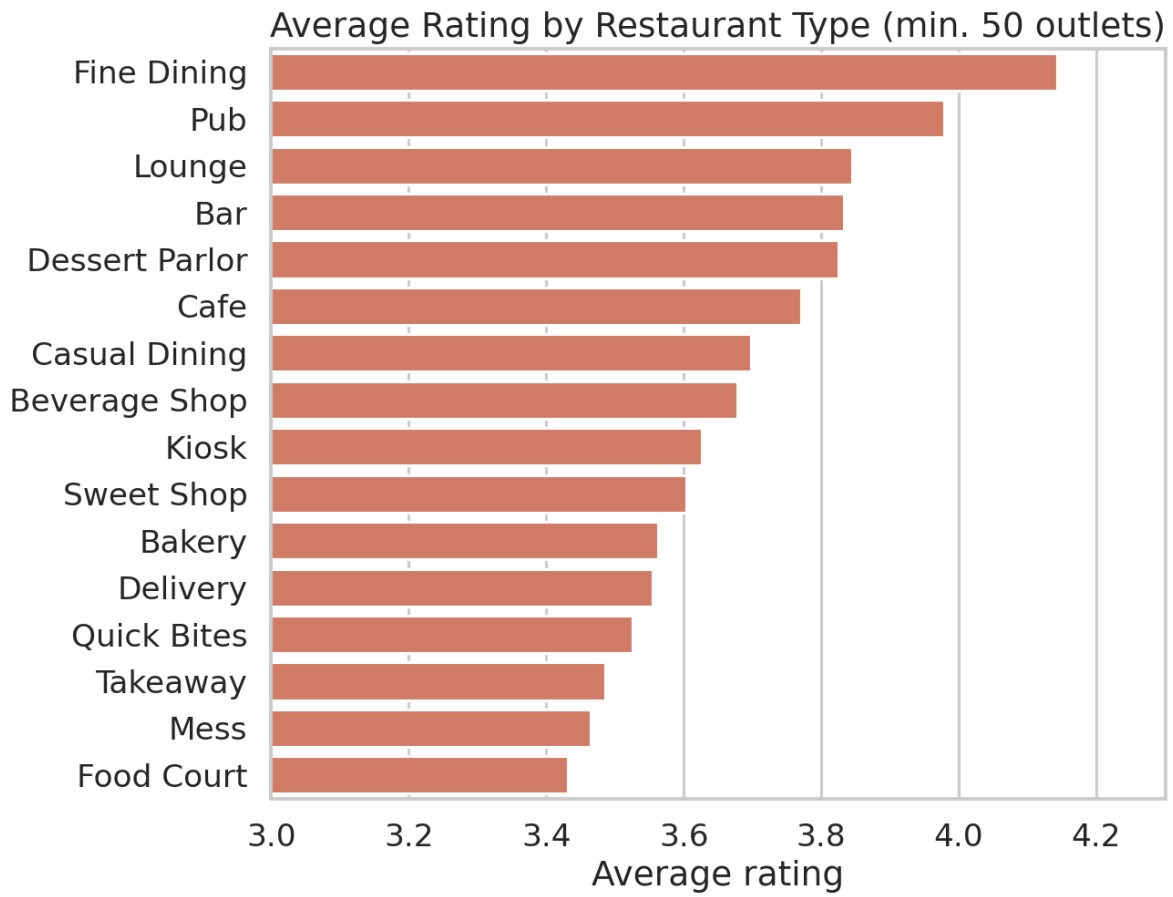
2.6 Service features: online ordering & table booking

Restaurants offering table booking average **4.07** vs. **3.57** for those that don't — the single largest gap found in this analysis. Online order availability shows a smaller, still-positive gap (3.64 vs. 3.58).

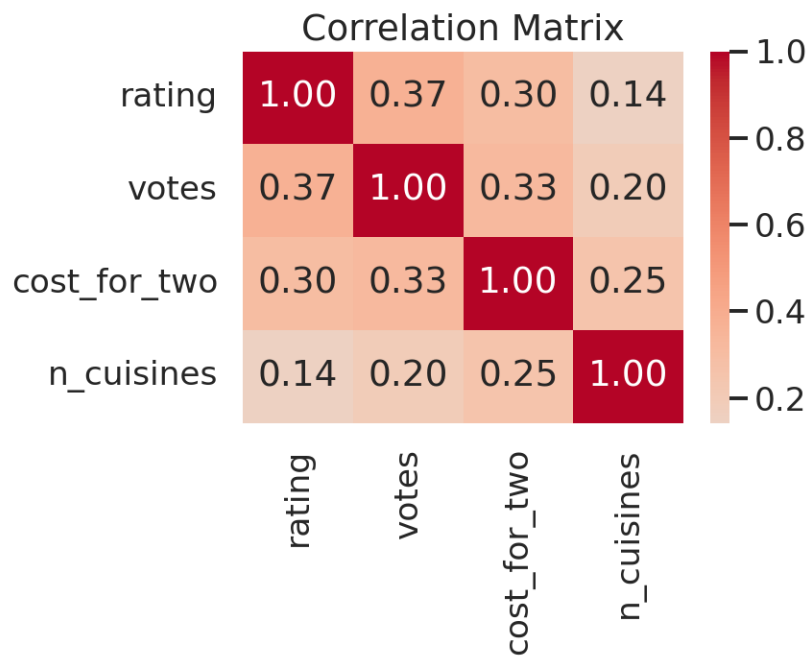


2.7 Restaurant format

Format is a stronger differentiator than cuisine alone: Fine Dining (4.14), Pubs (3.98) and Lounges (3.84) rate best; Food Courts (3.43), Mess (3.46) and Takeaway (3.49) rate lowest.



2.8 Correlation overview



Votes, cost, and rating are all mildly-to-moderately positively correlated; number of cuisines offered has little relationship with rating.

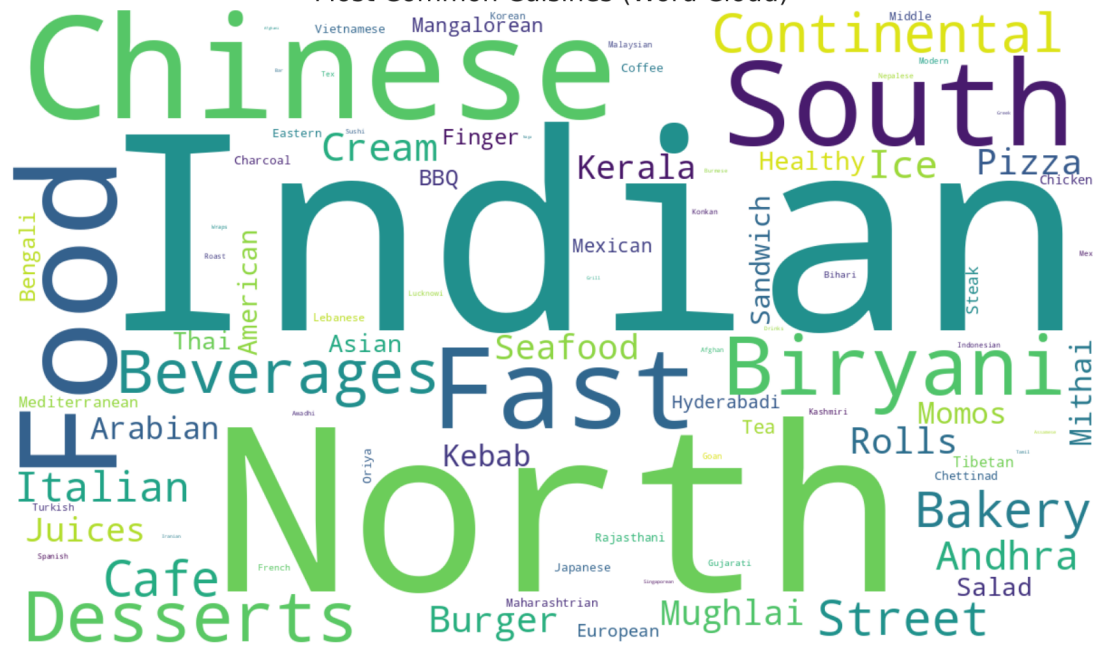
2.9 What are people ordering? (Word clouds)

Most Popular Dishes (Word Cloud)



Most frequently liked dishes across all restaurants.

Most Common Cuisines (Word Cloud)



Most common cuisine tags across all restaurants.

3. Key Findings

- **Data quality:** ~53% of raw rows were structurally corrupted (fields shifted by embedded review text) and had to be dropped; the cleaned dataset covers 10,296 unique restaurants.
- **Ratings skew positive but rarely excellent:** average 3.62/5, tightly clustered between 3.0–4.3.
- **Supply $\neq$ quality:** the highest-volume locations (Electronic City, Marathahalli) are not the highest-rated; premium central areas (Lavelle Road, Church Street, Koramangala) rate best.
- **Niche cuisines outperform staples:** Continental, Cafe, Desserts, Italian and Asian rate higher on average than North Indian, Fast Food and Biryani, despite far smaller restaurant counts.
- **Price correlates with rating** ($r \approx 0.30$) — restaurants above Rs.1,000 for two average well over 4.0.
- **Table booking is the strongest service-feature signal** — a +0.50 average rating gap versus restaurants without it.
- **Format matters more than cuisine** — Fine Dining, Pubs and Lounges rate highest; Quick Bites, Takeaway and Food Courts rate lowest.

4. Recommendations for an Alfido-Tech-style Platform

1. Target partnerships in high-quality, mid-supply micro-markets

Prioritise onboarding drives in areas like Koramangala (4th/5th/6th Block), Indiranagar and Jayanagar, which combine solid restaurant density with above-average ratings — a more efficient partnership ROI than oversaturated, mixed-quality zones like Electronic City.

2. Build a "Hidden High-Raters" discovery feature

Surface Continental, Asian, Italian, Cafe and Dessert concepts — cuisines that consistently out-rate the market despite much lower restaurant counts — giving users better choices and giving smaller/niche restaurants visibility disproportionate to their size.

3. Launch price-tiered content and curated collections

Since rating rises steadily with price bracket, replace generic "best restaurants" lists with budget-aware content ("Best under Rs.500", "Worth the splurge: Rs.1500+") so users across spending levels get genuinely well-matched, well-rated recommendations.

4. Incentivise table-booking and online-order adoption among partners

Table booking shows the single largest rating association in this dataset (+0.50 points on average). An onboarding programme that helps restaurants enable booking (and, to a lesser extent, online ordering) could lift marketplace-wide quality and user trust.

5. Operationalise the location × cuisine heatmap

Use heatmaps like the one in Section 2.5 to guide where to court new restaurant partners (e.g. better Continental/Cafe options in under-served but strong locations) and where to flag weak spots to users (e.g. Fast Food in Sarjapur Road and Electronic City rates poorly and may need quality-improvement outreach).

End of report — see accompanying Jupyter notebook (Zomato_Analysis.ipynb) for the full, reproducible code and cleaned dataset.